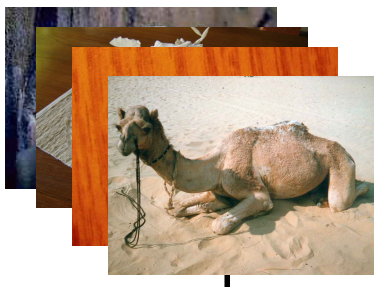
a. Incremental Training

New Class Images



Visual Foundation Encoder
(Frozen)

Feature Vector
$\vec{x} \in \mathbb{R}^d$

Sufficient Statistics

Class Mean
μ_c

Class Covariance Modeling
Σ_c

Number of Samples per class
n_c

No Backprop

No Replay

Gaussian Space

Class 1

Class 2

Class 3

New Class

b. Inference

Query Image



class

Visual Foundation Encoder
(Frozen)

Feature Vector
$\vec{x} \in \mathbb{R}^d$

MAP Classification

$$\arg\max_k p(c_k|\vec{x})$$

Class 1
$p(c_1|\vec{x})$

Class 2
$p(c_2|\vec{x})$

Class i
$p(c_i|\vec{x})$

Class n
$p(c_n|\vec{x})$

Predicted
$\hat{\text{class}}$